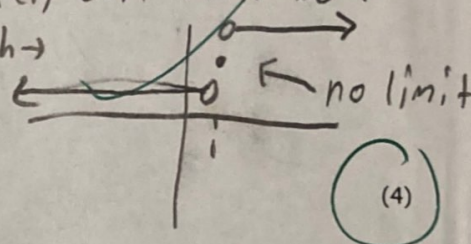


Total
20.5/21

1. Miguel believes that the $\lim_{x \rightarrow 1} f(x)$ exists if $f(1)$ is defined. Is there enough information for this statement to be true? Provide an explanation and sketch to support your answer

No, this is not true. This is because the limit does not care about the actual point, if $f(1)$ is defined, (2) the parts of the graph approaching $f(1)$ can still be in different places ($\lim_{x \rightarrow 1^-} f(x) \neq \lim_{x \rightarrow 1^+} f(x)$). Sketch →



2. Given the graph of $f(x)$, fill in the blanks with the correct answer.

a) $\lim_{x \rightarrow -4} f(x) =$

DNE

b) $\lim_{x \rightarrow 2} f(x) =$

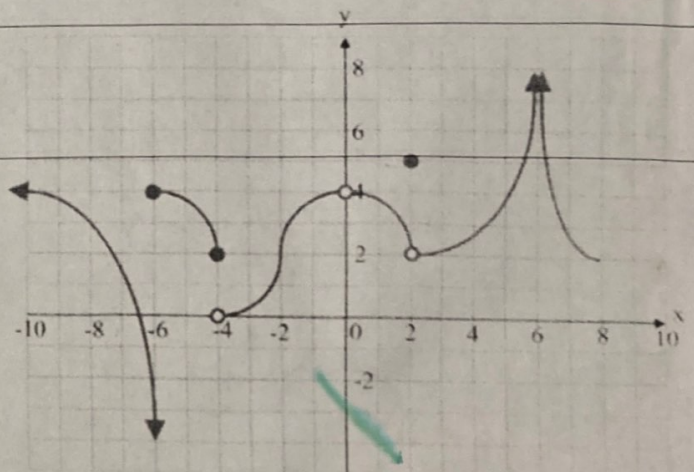
2

c) $\lim_{x \rightarrow -6} f(x) =$

4

d) $\lim_{x \rightarrow -6} f(x) =$

DNE



3. Sketch the following function $f(x) =$

$$\begin{cases} -x-2 & ; -2 \leq x \leq -1 \\ x & ; -1 < x < 1 \\ -x+2 & ; 1 < x \leq 2 \end{cases}$$

also shown by arrows on graph

- a) Graphically determine $\lim_{x \rightarrow -1} f(x)$

$\lim_{x \rightarrow -1^-} f(x) = -1$

$\lim_{x \rightarrow -1^+} f(x) = -1$

$\lim_{x \rightarrow -1} f(x) = -1$

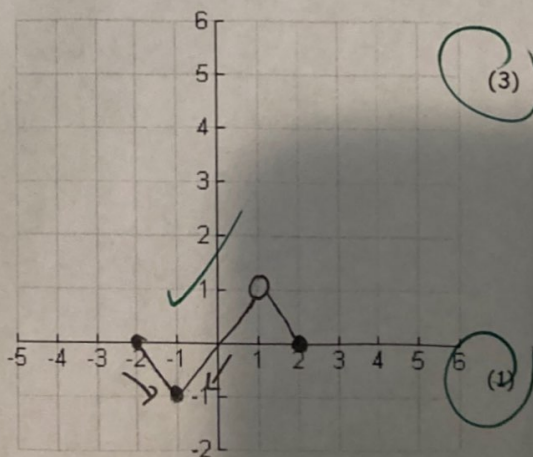
- b) Algebraically determine $\lim_{x \rightarrow 0} f(x)$

$-1 < 0 < 1$, use $f(x) = x$

$\lim_{x \rightarrow 0^-} f(x) = 0$

$\lim_{x \rightarrow 0^+} f(x) = 0$

$\lim_{x \rightarrow 0} f(x) = 0$



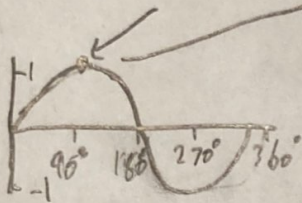
doesn't change
etc using
same function

4. Determine each of the following limits.

a) $\lim_{x \rightarrow 45^\circ} \sin(2x)$

$$\lim_{x \rightarrow 45^\circ} \sin(2x) = \sin(2(45^\circ))$$

$$= \sin(90^\circ) = 1$$



c) $\lim_{x \rightarrow -\frac{1}{2}} \frac{4x^2 - 1}{6x^2 - 5x - 4}$

$$= \lim_{x \rightarrow -\frac{1}{2}} \frac{(2x-1)(2x+1)}{(3x+4)(2x+1)}$$

$$= \lim_{x \rightarrow -\frac{1}{2}} \frac{2x-1}{3x+4}$$

$$= \frac{2(-\frac{1}{2}) - 1}{3(-\frac{1}{2}) + 4} = \frac{-2}{-\frac{11}{2}} = -2 \cdot \frac{2}{-11}$$

$$= \boxed{\frac{4}{11}}$$

b) $\lim_{x \rightarrow 2} \frac{\sqrt{9-x} - \sqrt{x+5}}{x-2}$

$$= \lim_{x \rightarrow 2} \frac{\sqrt{a} - \sqrt{b}}{x-2} = \lim_{x \rightarrow 2} \frac{a-b}{(x-2)(\sqrt{a} + \sqrt{b})}$$

$$= \lim_{x \rightarrow 2} \frac{9-x - (x+5)}{(x-2)(\sqrt{9-x} + \sqrt{x+5})}$$

$$= \lim_{x \rightarrow 2} \frac{-2(x-2)}{(x-2)(\sqrt{9-x} + \sqrt{x+5})}$$

$$= \frac{-2}{\sqrt{7} + \sqrt{7}} = \frac{-2}{2\sqrt{7}} = \boxed{\frac{-1}{\sqrt{7}}}$$

rationalize

d) $\lim_{x \rightarrow 81} \frac{x^4 - 3}{x - 81}$

$$= \lim_{a \rightarrow 3} \frac{a^4 - 81}{a - 3}$$

$$= \lim_{a \rightarrow 3} \frac{a^4 - 81}{(a^2 - 9)(a^2 + 9)}$$

$$= \lim_{a \rightarrow 3} \frac{a^4 - 81}{(a^2 + 9)(a-3)(a+3)}$$

$$= \lim_{a \rightarrow 3} \frac{1}{(a^2 + 9)(a+3)} = \frac{1}{(18)(6)} = \boxed{\frac{1}{108}}$$

5. Find the values of "a" and "b" so that the function

$$g(x) = \begin{cases} 3x+2a & ; x \leq -2 \rightarrow 3x \\ -\frac{1}{2}(x-2)^2 + 2b & ; -2 < x < 4 \text{ is } -\frac{1}{2}(x-2)^2 + 2b \\ \sqrt{x-4} - \frac{7}{2}a & ; x \geq 4 \rightarrow \sqrt{x-4} \end{cases}$$

continuous.

when $x = -2$

$$3b+2a = -\frac{1}{2}(-4)^2 + 2b$$

$$2a - b = 2b - 8$$

when $x = 4$

$$\frac{1}{2}(2)^2 + 2b = \sqrt{0} - \frac{7}{2}a$$

$$2b - 2 = -\frac{7a}{2}$$

$$a - 3 = b - 4$$

$$b = a + 1$$

$$2(b+1) = -\frac{7a}{2}$$

$$b+1 = -\frac{7a}{4}$$

$$b = -\frac{7a}{4} + 1$$

$$b = b$$

$$a+1 = -\frac{7a}{4} + 1$$

$$4a = -7a$$

$$11a = 0$$

$$\boxed{a = 0}$$

$$b = a+1$$

$$b = 0+1$$

$$\boxed{b = 1}$$

check

$$3(-2) + 2(0) = -6$$

$$-\frac{1}{2}(-4)^2 + 2(1) = -8 + 2 = -6$$

$$-\frac{1}{2}(2)^2 + 2(1) = -2 + 2 = 0$$

$$\sqrt{4-4} - \frac{7}{2}(0) = 0$$